

2D SCANNING LIDAR · LR-1BS SERIES

ZONE-SWITCHING OLEI LR-1BS2 270° LIDAR

A 450-gram 270° scanner that does the zone logic itself — 16 banks of three protection levels, four switching inputs and four switching outputs — so a simple platform can stop for an obstacle without a PLC in the loop.



■ WHY THIS SENSOR

- **270° field of view from one corner mount** — a single sensor covers the front and both flanks of an AMR or a machine envelope
- **16 banks × 3 protection levels** — warn, slow and stop zones drawn per bank and switched live from the platform
- **4 switching inputs / 4 switching outputs** — select the active bank and drive the stop signal directly, with or without a PLC
- **Point cloud *and* switching output together** — azimuth, distance and signal strength over Ethernet while the relay logic keeps running
- **0.225° angle resolution at 10–25 Hz** — 16K–40K points per second, 130 ms response at 10 Hz
- **≤ ±1° horizontal plane error** — the scan plane stays where the drawing says it does
- **5 W at 12–24 V DC, IP65** — runs off platform power and survives a wash-down bay
- **Oforge zone configuration, OLAMViewer and a ROS driver** — set the banks in the GUI, stream the cloud into an existing stack

0.2–10 m

WORKING RANGE

270°

FIELD OF VIEW

16 × 3

ZONES / LEVELS

4 / 4

INPUTS / OUTPUTS

■ APPLICATION SCENARIOS



AMR & AGV

270° obstacle avoidance from a single chassis-mounted scanner



INTRALOGISTICS

Forklifts, tuggers and conveyor zones in active aisles



AREA MONITORING

Perimeter and equipment-envelope intrusion detection



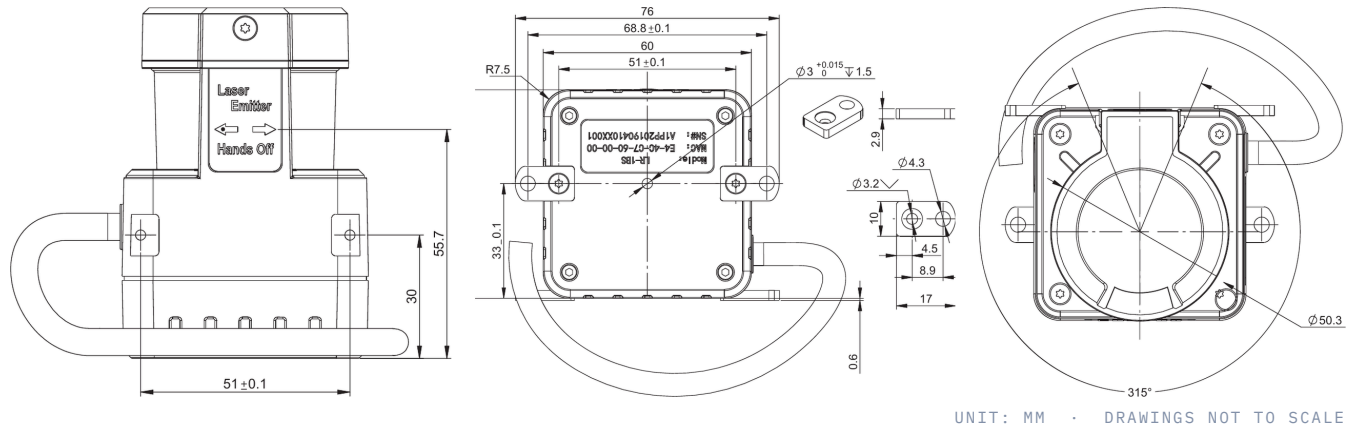
INDUSTRIAL CELLS

Machine-envelope monitoring and presence sensing

PARAMETER	TECHNICAL DATA
Sensing technology	2D scanning ToF LiDAR
Laser wavelength	905 ± 20 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Laser channels	1
Field of view (horizontal)	270°
Beam divergence	5.8 mrad longitudinal 0.58 mrad transversal
Angle resolution (horizontal)	0.225°
Scanning frequency	10–25 Hz
Working distance (based on reflectivity)	0.2–10 m @ 80% reflectivity 0.2–8 m @ 10% · 0.2–2 m @ 4% · 0.2–10 m @ reflector
Accuracy ①	Absolute < ±50 mm Repeat < 20 mm
Distance resolution	1 mm
Signal strength	0–60000
Scanning rate	16K–40K pts/s
Horizontal error (horizontal angle)	≤ ±1°
Response time	130 ms @ 10 Hz

PARAMETER	TECHNICAL DATA
Number of bank fields	16 banks, 3 levels/bank
Switching inputs	4
Switching outputs	4
Ethernet connection	100 Mbps Ethernet
Connection	12-core cable, 2 m 4-pin M8×1 A-code female, 2 m LAN
Operating voltage	12–24 V DC
Power consumption @ 25 °C	5 W
Mechanical dimensions	≤ 60 × 60 × 85 mm
Weight	450 g
Degree of protection	IP65 or optional
Operating temperature	–10 to +50 °C
Storage temperature	–30 to +70 °C
Ambient light limit	80 000 lux
Indicator status	RGB × 4
Mounting	Bottom / back
Function software	Oforge
Basic software	OLAMViewer · ROS driver

■ OUTLINE AND MOUNTING



Certification. FCC, EMC, CE, RoHS, REACH, FDA. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. ① Standard value; actual value depends on environmental conditions. Accuracy and working distance both degrade with lower target reflectivity — size the zones from the 4% and 10% figures, not the 80% figure.

Installation. Mount the scanner so the 270° plane is unobstructed through its full sweep and the blind 90° faces structure, not traffic. Keep the optical window clear of spray and dust, clean it on a schedule in dirty environments, and verify every configured bank on the platform after any change to mounting height or angle.

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